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The Role of FinTech in Enhancing Financial Inclusion for SMEs in Emerging Markets

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ABSTRACT: This article examines the transformative role of Financial Technology (FinTech) in promoting financial inclusion for small and medium-sized enterprises (SMEs) across emerging markets. SMEs, which form the backbone of these economies, often face structural barriers such as lack of collateral, high borrowing costs, and limited access to formal financial services. FinTech solutions offer innovative ways to address these gaps by lowering transaction costs, expanding credit access, and improving financial literacy. Through case studies from Africa, Asia, and Eastern Europe, the analysis demonstrates how FinTech fosters entrepreneurship, supports women-led businesses, and enhances resilience against financial exclusion. However, challenges such as regulatory constraints, cybersecurity risks, and gender-blind solutions remain significant. The article concludes that a balanced regulatory framework, combined with public–private partnerships and targeted digital literacy initiatives, is crucial to fully harness FinTech's potential to drive inclusive and sustainable SME growth in emerging economies.

INTRODUCTION

Financial inclusion enhances access to affordable financial products and services to underserved individuals and firms. In particular, the availability of credit is crucial to start and grow businesses. The widespread use of digital technologies enables new forms of information production and processing. FinTech firms embrace these possibilities, challenge incumbent financial institutions (FIs), and offer financial products to underbanked segments. The extent to which FinTech is already driving inclusion remains, however, unknown. This study investigates the financial

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challenges faced by underserved entrepreneurs; FinTech credit solutions that carefully define entrepreneurial quality; and the impact of FinTech on new and underserved entrepreneurs. FinTech lending matches the investment needs of young businesses, decreases credit denial, and targets undersupply categories like start-ups without collateral, establishing a channel for financial and social inclusion. At the country level, access to FinTech credit correlates with a larger share of new businesses (Isabel Pavón Cuéllar, 2018). These findings inform policy choices regarding the regulation of digital credit markets and data sharing.

1. UNDERSTANDING FINANCIAL INCLUSION

Financial inclusion entails universal access to affordable and appropriate financial products and services provided by formal financial institutions, enabling individuals and enterprises to manage financial matters, achieve economic goals, and mitigate risks (Mhlanga, 2024). Many emerging economies have achieved substantial inclusion in savings and loan products, yet barriers persist. Limited access to credit is a particular obstacle, especially for women and SMEs, with previously excluded groups encountering the most significant impediments (Isabel Pavón Cuéllar, 2018). Enhancing financial services for underserved segments constitutes a major objective. Financial innovation utilizing information and communication technology (ICT) introduces transformative opportunities to extend coverage and reduce the costs of financial access.

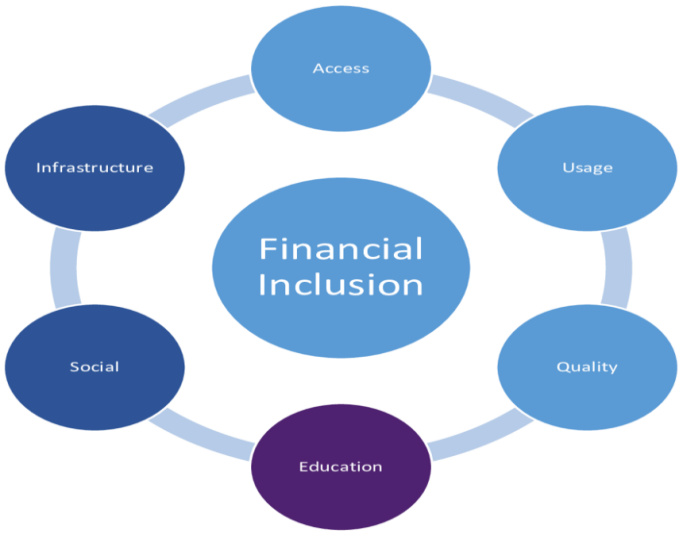
1.1. Definition and Importance

Financial inclusion is defined as the extent to which formal financial services are available and accessible to the unbanked and under-banked (Isabel Pavón Cuéllar, 2018). Access to finance facilitates the development of individuals, businesses, and nations at large, and therefore access to a broad set of appropriate financial services is recognized worldwide as a key contributor to economic growth and poverty reduction. However, establishing an exact measure for financial inclusion can be challenging. The establishment of new financial institutions in a country generally tends to increase the access of financial services to households and SMEs and thus financial inclusion. As banks are the most dominant providers of deposit accounts, loans, and other financial services, all these services are expected to positively increase with additional banking institutions. The generalized access to financial services is a promising source of growth and social inclusion in the quest to reach a “decent life for all”. Some groups, such as the poor, women, youngsters, and inhabitants of rural communities, face more barriers to access these services. Recently created and small businesses also encounter numerous obstacles. Achieving more bank borrowers depends on access channels, product attributes, and the socio-economic characteristics of the potential clients. Analyzing the components of supply and demand can build adequate models to statistically validate the relative importance and robustness of the factors that affect the use of financial services.

1.2. Challenges in Financial Inclusion

Despite the positive role of SMEs in economic development and their potential to alleviate unemployment, their growth and contribution in emerging markets remain limited. Ownership and management of informal credit sources, often tenancy of ordinary banks, abundance of principles in formal and informal financing and development, inward policies of banks, fluctuation in economic activities, abuse of financial provisions, and lack of adequate government assistance have been considered as factors limiting SMEs in emerging markets. Small and Medium Enterprises (SMEs) lack access to adequate external long-term finance, which constitutes a significant obstacle to their growth. This situation of limited access may be seen in the context of a financial system which is generally characterized by under-development and shortages of capital in low-income countries. When compared with large firms, SMEs face various specific difficulties, such as informational opacity and the priority of reputation and flexibility. Basically, SMEs and potential entrepreneurs in emerging markets encounter difficulties and challenges related to obtaining information, accessing to funds, acquiring financial support, and getting access to technical assistance.

Figure 1 : Six elements of financial inclusion



Source : Angela C. Lyons et al. (2017)

Mrs. Bieke Lemahieu, Technology Strategy Analyst at Abi, stated that one of the important challenges fintech startups face is the complicated and unreliable process of opening current accounts. Furthermore, other challenges such as compliance with Know Your Customer (KYC) and Anti-Money Laundering (AML) standards, invisible and intransparent regulatory guardrails and often nonexistent API infrastructure also discourage entrepreneurs from embarking on this niche. Even those startups which do manage to build a minimum viable product face a tedious onboarding process. Such processes result in the loss of potential SME customers.

The most important impediment for informal firms in Africa which want to access external finance and formalise is related to structural and institutional problems rather than shortcoming of the lack in financial package of the formal banking sector. In the general quest to provide a more conducive environment to facilitate financial access by the informal sector further serious analysis of development of a comprehensive institution which focus on structural problems of institutions is needed by actors and policy makers (Woldie et al., 2018).

2. OVERVIEW OF FINTECH

Financial technology (FinTech) has been transforming the financial industry since the late 2000s through a combination of social, economic, and institutional factors (Rahman Aleemi et al., 2023). Global banking markets have stretched regulatory capacity, shrinking profit margins. Meanwhile, a generation disinterested in finance seeks modern services, and ensuring financial inclusion is a priority for the United Nations, the World Bank, and other organizations. Large digital technology firms have entered financial services, displacing incumbents and prompting reform through open banking principles. Further, blockchain technology challenges the intermediated model by allowing people to trade currency, stock, contracts, and other instruments directly.

FinTech thus offers a prescription for a more inclusive, dynamic, and competitive financial industry. Even small marketshare firms can reshape the national and global financial infrastructure, disrupting banking services that previously benefited from scale restrictions and barriers to entry. FinTech capabilities include payments, e-wallets, mobile banking, money transfers, and personal capital management. Micro-loans are supplied by numerous fintechs and larger firms operating in the sector, underwritten and approved by machine learning algorithms trained with big data from alternative sources. Crypto-assets perform various financial sector roles from borrowing and lending to transfers and asset purchase.

Financial inclusion has been a priority in African countries for at least a decade, supported by initiatives to promote mobile payments and increase access to loans. Additional support derives from regional economic communities like the Common Market for Eastern and Southern Africa (COMESA). FinTech for Development agenda enables low-income persons to be included in the digital economy through financial transactions and capabilities in advance of a more complete digital identity and other infrastructure. Credit and micropayment facilities address opportunities for underserved regions, reminding of the micro-credit initiatives popularized by the Grameen bank. Micro-insurance and social investment platforms create low-cost alternatives and help mobilize capital. Finally, specialized financial solutions improve the economic situations of women and SMEs more generally.

2.1. Definition and Evolution of FinTech

Many theories and definitions of FinTech exist, ranging from broad concepts encompassing the entire financial and technological infrastructure to narrower descriptions focused on specific technologies (Leong & Sung, 2018). FinTech stands for "financial technology." The Association

for Financial Professionals defines FinTech as technologies applied to financial services that aim to enhance financial activities. From a marketing perspective, FinTech can be defined as the integration of technology into offerings by financial services companies to improve their use and delivery to consumers. Several core technologies and strategic capabilities fuel FinTech innovation, with startups becoming essential as a significant source of technological advancement and innovation in the banking sector.

2.2. Key Technologies in FinTech

FinTech represents the fusion of finance and technology, and is defined as the innovative application of technology in the design and delivery of financial services and products. By combining financial services, computer science, and information systems, a new financial ecosystem has emerged, providing users with diverse services such as payments, insurance, investments, and financing in innovative and creative ways. FinTech offers safe and secure financial services through online platforms, mobile technology, big data, and advanced analytics, and has the potential to address challenges to financial inclusion and reduce financial exclusion. New digital financial services capitalise on cloud computing, big data, and machine learning, concealing complexity and infrastructure to 'democratise' access to financial services (Mhlanga, 2024). Lower transaction costs enable mass customisation and have extended reach, the potential to relax financial constraints (with more accessible credit and improved information), investment in human capital and entrepreneurship, and greater resilience to adverse shocks. FinTech services provide digital access to credit, savings, insurance, and financial support for SMEs, as well as lay the foundation for transport and logistics, telecommunications and digital payments systems, infrastructure, and election conduct.

3. THE LANDSCAPE OF SMES IN EMERGING MARKETS

Small and Medium Enterprises (SMEs) constitute a significant portion of the overall economy (Mncedi Mahlati, 2018). The existing variety of financial products and services is not adequate to meet the needs of small businesses (Elasrag, 2016). A substantial number of SMEs prefer banks to transition from product pushers to holistic advisors. Effective risk management strategies, supply chain management skills, and knowledge reduce perceived SME risks. Banks often have capable business managers who can provide sound advice and maintain strong client relationships. A digital ecosystem influences banks' ability to meet the needs of small companies and enhances customer experience and loyalty. Partnering with Fin-Techs assists banks in efficiently engaging with the SME market and capturing revenue. SMEs require support to obtain more business with simplicity, reliability, and speed. Banks should consider an ecosystem approach to address the varied needs of SME customers and establish dedicated departments for small clients.

SMEs make up the economic tissue of many countries. In developing nations, they represent the majority of employment, including female employment. Globally, SMEs constitute the overwhelming majority of firms, exceeding 95% of all businesses and accounting for approximately 50% of GDP and 60%–70% of total employment. Between 420 million and 510

million SMEs exist worldwide, with 310 million located in emerging markets. Investing in SMEs constitutes a strategic move with sustainable returns that benefit regions, countries, and societies. Islamic finance offers development and growth opportunities for SMEs but also presents challenges; Islamic financial institutions are positioned to support SME expansion.

3.1. Characteristics of SMEs

Small and Medium Enterprises (SMEs) represent the backbone of the emerging market economy, with more than 40 million SMEs estimated to be active across emerging-market economies (Elasrag, 2016). Worldwide, SMEs constitute about 95 per cent of all firms, contribute approximately 50 per cent of GDP and provide 60–70 per cent of total employment. Within these countries, 75 per cent of industrial jobs, but only 11 per cent of capital expenditure, can be assigned to the SME segment (Mncedi Mahlati, 2018). Addressing the financing gaps encountered by SMEs therefore offers tremendous opportunities for FinTech companies, and many financial services providers classify this segment amongst their priority growth markets.

SMEs are characterised by a number of defining features. In terms of organisational form, SMEs range from informal microenterprises and family businesses to formal established firms. Legal and organisational forms of SMEs include sole-proprietorships, partnerships (both limited and unlimited), and privately owned corporations. Younger firms often tend to be smaller; however, there is no clear evidence on whether businesses start small and grow or start at different sizes and follow various growth paths. The ownership is mostly private and managerial, and less than 30 per cent of companies are publicly listed. Asset operations are less than $15 million and fewer than 500 employees are engaged. SMEs tend to be labour-intensive, employing mostly unskilled workers, more so in smaller firms. In terms of economic contribution, firms with 0–19 employees contribute approximately 45 per cent to total value added, while firms in the size range of 20–99 employees account for 23 per cent, with the rest coming from more than 100 employee firms.

Recent experience shows that SMEs in emerging markets confront specific constraints. Important supply-side restrictions to SME financing include asymmetric information and the absence of collateral due to size and operational restrictions. These difficulties are compounded by low operating margins, poor corporate governance, limited operating histories, and a lack of audited financial information. SMEs in emerging markets typically operate in the informal economy, which further constrains their access to financing. Economic uncertainties, lack of risk capital and inefficient judicial systems exacerbate the situation. On the demand side, the inadequacy or limited awareness of entrepreneurial skills and general business training impede the development of bankable proposals. While the overall demand for financing is high, credit will be sought only if the expected returns from investment are sufficient to compensate for the cost of credit. SMEs often have lower profit margin projects, and the investment proposal is likely to create additional credit needs that the SME may be unable to service.

3.2. Barriers Faced by SMEs

SMEs face significant obstacles including limited access to finance, lack of access to technology, regulatory hurdles, and low financial literacy (Woldie et al., 2018). Accessing finance is a well-recognized barrier to SME growth in emerging markets, where FinTech solutions can exert a transformative impact (Mncedi Mahlati, 2018). In many African nations, stringent collateral requirements and high-interest rates exclude a large share of credit-seeking small businesses; furthermore, as observed in the DRC, many financial institutions erroneously perceive women-led enterprises as nonviable, reflecting broader societal norms that constrain women's economic participation. Regulatory hurdles are manifold in many emerging-country contexts. Effective cooperation between regulators and firms to tackle these constraints remains elusive. Firms argue that many regulatory demands are irrelevant to their operations, burdensome to implement, and increase operating costs. A further obstacle is that many SMEs have limited exposure to digital technologies, constraining their ability to engage with FinTech products and services; lower levels of digital adoption result in restricted use of e-commerce channels for sales and purchases, diminished information exchange capabilities, and reduced prospects for digital transformation and innovation. These obstacles highlight the importance of policies aimed at closing the financial, digital, and skills divides to place more SMEs in a transformative position and ensure inclusive, sustainable economic development; they also set the agenda for intervention by FinTech firms, which can generate inclusive development by extending access to underserved SMEs and addressing other constraints.

4. THE INTERSECTION OF FINTECH AND SMES

SMEs are an important segment for financial institutions, with banks perceiving them as profitable clients. Bank financing significantly enhances SME profitability, incentivizing greater engagement from banking institutions. Despite this, SME-specific challenges continue to restrict access to bank finance. Mitigating these constraints demands concerted efforts from all relevant stakeholders, supported by policies that foster trust between financial institutions and SMEs. Technology has become a crucial competitive tool for driving operational excellence and delivering superior service quality. In response, banks are pursuing innovative means to establish personalized and constructive relationships through digital channels. Strategic partnerships occupy a central role in banks' digital transformation strategies. Empirical studies indicate that a high-touch relationship model is the most appropriate for serving SME customers, balancing economic considerations with best practices; consequently, assigning a relationship manager to each SME client is justified by market size. Advanced segmentation approaches based on lifestyle, business lifecycle, and channel preferences can unlock additional revenue potential. Banks possess experienced business managers who provide sound advice and insight to SMEs; moreover, the adoption of effective risk management strategies, people management practices, and supply chain skills by SMEs reduces perceived market risk. Participation in a digital ecosystem enhances banks' ability to meet SMEs' unique needs. Collaborations with Fin-Tech firms offer a more efficient means to engage the SME market, thereby helping traditional banks capture its revenue. SME

clients require support to obtain more business with simplicity, reliability, and speed, prompting recommendations for an ecosystem approach to address diverse needs. Establishing a standalone, specialist SME department is advisable to deliver tailored services (Yeboah et al., 2014). Some Fin-Tech solutions have been designed specifically for SMEs, outside of traditional banking channels. For example, M-Pesa, founded in 2007, is a mobile money service in Kenya that allows customers to store money in a mobile wallet and make cashless transactions. Branch, a Fin-Tech lending platform, uses mobile phone data to assess individual creditworthiness and provides short-term loans to underserved populations without a credit history. MAVO aims to unlock access to capital for African SMEs by extending credit to employees of small enterprises and loaning an equivalent amount to the SME itself. These platforms illustrate how financial technology can offer alternative pathways to finance and payment mechanisms, circumventing the barriers inherent in conventional banking systems .

4.1. FinTech Solutions for SMEs

The digitalization of the financial sector has produced new opportunities for micro, small, and medium enterprises (SMEs), supporting their growth and development. Data collated from 103 countries during 2014–2017 provides robust evidence that the adoption of financial technology (FinTech) firms accelerated SME external financing, particularly in countries with an underdeveloped banking system, weaker institutional environments, and a lower income level. Small-and-medium-size enterprises (SMEs) contribute 50% of gross domestic product (GDP) and create 60% of jobs in emerging countries, yet only 20% of them have access to formal financial services, necessitating alternative solutions. The digitalization of finance with FinTech creates new opportunities for SMEs. Drawing on the European experience and the case of Ukraine, this study aims to explore the role of FinTech in enhancing SME financial inclusion.

FinTech, the intersection of finance and information technology, has transformed the global financial services system. Core FinTech technologies, including cloud computing, big data analytics, Artificial Intelligence (AI), blockchain, Application Programming Interfaces (API), and mobile technology, have facilitated the delivery of innovative financial products and services that directly or indirectly improve financial inclusion. In particular, FinTech has transformed payment systems, enabling the rapid transfer of funds and increased access to payments for individuals, thereby improving credit availability through online lending platforms. In Ukraine, FinTech platforms enhance SME access to finance by digitalizing credit granting, reducing informational asymmetry between borrowers and lenders, and facilitating the use of e-money. Despite regulatory challenges, local FinTech start-ups have succeeded in deploying such solutions, illustrating their potential as vital instruments for SME finance (Ivashchenko et al., 2018).

4.2. Case Studies of Successful FinTech Implementations

Digital technologies facilitate information sharing and connectivity among financial entities. Digital finance reduces information and transaction costs, thereby improving financial services and access. Small and medium-sized enterprises contribute over 50 percent of China's tax revenue,

60 percent of gross domestic product, and 90 percent of companies; they represent a vital engine of economic growth and employment. Nonetheless, SMEs face longstanding constraints on finance, owing to limited resilience to risk and comparatively short cash reserves, a reality exposed during times of crisis. Recognising the contribution of SMEs, and that of the service sector broadly, a digital big-data platform has been developed to bridge information asymmetries and help firms obtain finance. More-tailored support therefore exists to help SMEs overcome financing constraints, although measures such as interest-rate subsidies continue to crowd out private-sector solutions

Regulators are increasingly developing new approaches to tackle FinTech, as existing regimes oftentimes do not support innovative products and business models. Distributed ledger technology, for example, may be used to change or replace the overall setup of market participants and infrastructures, thereby creating new risks. The growth of online lending platforms, particularly in China, has highlighted the need for regulatory intervention to mitigate fraud and mismanagement. The case of Ezubao, a peer-to-peer lending platform that operated as a Ponzi scheme and collected approximately USD 9 billion from over 900,000 investors, exemplifies the dangers of insufficient oversight (Didenko, 2018).

5. IMPACT OF FINTECH ON FINANCIAL INCLUSION

The mobilization of financial resources through technologically-enhanced delivery channels improves the flow of capital from providers to users by reducing transaction costs, addressing information asymmetries, and facilitating digital on-boarding. The increased availability of cost-effective client information supports broad-based and inclusive access to capital for the real economy, especially small businesses, and the design of new financial products adapted to local needs. Non-traditional financial service providers leverage data sets derived from users' online activities and interactions with their devices in support of credit underwriting and customer acquisition; they also exploit different modes of delivery to enhance cost efficiency and otherwise improve the experience of their customers. Building on such activities and capitalizing on their technical, organizational, and commercial capabilities, they extend their operations by entering new markets, broadening the spectrum of activities, and strengthening their role in the financial system. The use of digital financial services can impact financial inclusion by lowering accessibility thresholds or providing under-served customers with a first-time access to the formal financial system.

5.1. Access to Capital

The provision of funding to SMEs is one of the more widely recognized benefits of FinTech in the literature. These enterprises are often the most financially excluded, especially across emerging markets and developing countries (Isabel Pavón Cuéllar, 2018). Partly, inadequate access is a result of imperfect borrower information and incapacity of financial institutions to fund unsecured loans: the larger relative operating costs associated with lending to smaller customers also represents a deterrent. The shortfall of working capital in emerging economies represents an

additional obstacle for SMEs, as underscored by the disproportionate share of firms identified as financially constrained in cleaning services, retail, and wholesale sectors.

Societal and cultural constraints may further limit borrowing, and requiring women-owned micro- and small enterprises (MSEs) to obtain cosignatories constitutes a key barrier across Sub-Saharan African (SSA) countries. Importantly, credit market segmentation based on gender increases the weight of the borrowing burden on women holders. Adopting risk-based KYC and broadening the range of data types inclusively incorporated in credit reference bureaus have proved among the most effective measures to address these gaps. Certain supply-side inadequacies characterize this scenario: many private institutions misjudge low-income market opportunities because of a paucity of consumer insights. A deliberate commitment to incorporating additional informational data sets in the mix represents an essential first step to clearly improve understanding of the underlying drivers informing consumers' behavior. FinTech companies that provide credit consequently encounter difficulties in raising capital for growth; donors cannot solve these funding constraints directly, but may mitigate them through grants, impact investments, guarantees, and technical assistance. Mobile financial services and other FinTech innovations have enhanced women's households' decision-making, financial resilience, and economic prospects novel FinTech have indeed the potential to lift rural communities in SSA above the poverty line. Nevertheless, solutions remain largely "gender-blind" and are yet to be specifically tailored to the particular needs of women in SSA .

5.2. Improved Financial Literacy

The influx of FinTech solutions has been instrumental in enhancing financial literacy among small and medium enterprises (SMEs), particularly in emerging markets . By enabling SMEs to comprehend and make use of financial services, FinTech platforms provoke demand for formal financial products and services, thereby encouraging more enterprises to report their business activities and financials. This generates valuable alternative credit data for established financial institutions, which mitigates their reluctance to lend to SMEs lacking historical credit records. Beyond improving waste reduction, the FinTech ecosystem fosters entrepreneurship and innovation by addressing information and communication gaps between SMEs and potential lenders. Continuous interaction on FinTech platforms inculcates financial discipline and literacy among enterprises, further nurturing entrepreneurial capabilities and reducing transaction costs associated with loan applications from formal financial sources.

At the firm level, entrepreneurs equipped with financial knowledge exhibit better firm performance when financial capital is available (Brownhilder Ngek, 2016). The absence of adequate financing impedes the translation of financial literacy into improved SME outcomes. Therefore, entrepreneurs should not only enhance their financial knowledge but also develop effective means to access financial resources. Policymakers are encouraged to bridge the financial gap and create an enabling environment that facilitates affordable and accessible financing for entrepreneurs.

5.3. Cost Reduction in Financial Services

Digitization not only supports access to finance but also contributes to financial conduct and reduces the costs associated with providing and receiving financial services (Rahman Aleemi et al., 2023). Digitization offers an appealing approach to reduce the cost of providing financial services, which is crucial for inclusive finance and poverty alleviation. The continual digitization of the financial sector also benefits the conduct of financial institutions by enabling the collection of vast amounts of data from diverse information sources. Information technology and credit analytics techniques can be used on this data to build sophisticated credit risk models to better assess the riskiness of potential borrowers, especially those without traditional credit profiles. Although the digital transformation offers numerous advantages to both financial institutions and customers, it also presents new risks, including the risk of cyber-attacks and fraud. Virtual banking provides multiple advantages to financial institutions and brings benefits for customers, yet large gaps remain before low-income earners and small businesses become customers of virtual banks (Boshkov, 2019).

6. REGULATORY CHALLENGES AND CONSIDERATIONS

For the unbanked and financially excluded, FinTech offers financial inclusion by providing affordable services customized to the specific socioeconomic and cultural context. The success of these initiatives depends on the resilience of regulatory frameworks and how they mitigate the risk of data privacy and fraud (Didenko, 2018). The dominant model of regulatory compliance remains the command and control approach. Technology never works smoothly. Many of the firms that the FinTech industry aims to disrupt remain major actors under the current model. Yet, the evolving use of technology means that various mechanisms and choices of market behaviour can emerge rapidly, increase and decrease in sectors, and switch in and out of financial products and services far faster than regulators are able to contemplate (Barberis & Ross P. Buckley Douglas W. Arner, 2017).

6.1. Regulatory Frameworks in Emerging Markets

Emerging markets host many tens of millions of micro, small and medium-sized enterprises (SMEs). They lack access to finance and rely on their professional networks and retained earnings. FinTech holds out the promise of substantially expanding access, and delivering complementary benefits by lowering the cost of servicing customers, typically towards the end of the long-tail of small businesses. During the beginning of the twenty-first century, FinTech acquired momentum in developed countries where it was initially supplemented by one or more major industrial Internet platforms and an infrastructure of inter-operability. The same cannot be said about emerging-market jurisdictions, except for Kenya where a wealth of both research and development insights are available. Regulatory frameworks for FinTech at the emerging-market level are subject to considerable heterogeneity and ongoing experimentation, with very substantial challenges in areas such as data protection and privacy, record keeping, and regulatory forbearance (Didenko, 2018).

6.2. Balancing Innovation and Regulation

Achieving a balanced regulatory environment for FinTech remains a considerable challenge. Fostering innovation and enabling broader access to financial services need to be reconciled with established policy objectives such as consumer protection, as well as broader concerns related to systemic stability, cyber-security and financial crime. Existing regulatory regimes generally do not support new FinTech products and business models, rendering it difficult for regulators to weigh the benefits of financial innovation against associated risks. At the same time, approaches that are excessively permissive risk undermining financial stability and technological progress. Distributed ledger technology offers a relevant case in point, since it departs fundamentally from existing market structures by eliminating traditional counterparts and intermediaries. Yet for all their transformative potential, these systems also raise new risks that can be difficult to monitor and contain (Didenko, 2018). Online lending platforms provide useful lessons concerning the consequences of inadequate regulatory safeguards, given that their rapid growth reflected strong demand but one that was not matched by appropriate oversight. The collapse of Ezubao – a Ponzi scheme that collected nearly USD 9 billion on a peer-to-peer platform – pointed to vulnerabilities arising from gaps in the regulatory framework and highlighted the importance of effective governance and transparency mechanisms. FinTech products and services also raise cross-cutting concerns that lie beyond the conventional perimeter of financial regulation. Procedures for implementing the Anti-Money Laundering (AML) and Combating the Financing of Terrorism (CFT) standards encounter a number of inherent constraints in the context of FinTech activity . At the same time, standards relating to privacy and data protection can, if misapplied, impede the flow of information across financial counterparts and jurisdictions. Decentralised solutions dating to the earliest stages of financial innovation pose a different challenge all together, since the very nature of this activity may preclude direct oversight and require a more synthetic approach focused on the ecosystem as a whole. Together, these observations suggest the scope and complexity of regulatory responses to emerging FinTech products and services. Despite their focus on the financial technology sector, many regulators still struggle to address the diverse range of new business models and enable the industry and its customers to benefit from them. At the same time, authorities must continue to invest in capacity-building programs and the development of regulatory toolkits.

7. RECOMMENDATIONS FOR POLICYMAKERS

Policymakers have an important role to play in fostering FinTech development and adoption. Regulatory frameworks need to balance innovation and oversight to encourage FinTech growth without stifling competition and inclusiveness. Public–private partnerships can mobilise expertise and resources to develop targeted solutions for SMEs and other excluded groups.

Given the limited understanding of the financing gap faced by SMEs, undertakings to map SME credit constraints and their sectoral composition are still required (Ivashchenko et al., 2018). FinTech cannot by itself close the entire gap. Conversely, it can be a pivotal tool to mitigate the

effects of the recessions brought about by the COVID-19 pandemic on emerging SMEs. A better understanding of bank SME credit activities in emerging regions and how FinTech can aid those activities is required before concrete conclusions can be drawn (Woldie et al., 2018).

7.1. Creating Supportive Regulatory Environments

Enabling FinTech solutions to flourish necessitates the establishment of an appropriately tailored regulatory framework (Ivashchenko et al., 2018). Regulatory authorities are challenged to devise approaches that strike a balance between the pursuit of economic growth and the safeguarding of individual and contributory entity interests, all the while preserving public confidence and preventing systemic repercussions. Options open to regulators include crafting distinct regimes for each new product, updating and revising standing regulation, or instituting a unified regime covering multiple products (Didenko, 2018). Since newly promulgated rules may be difficult to predict, a principles-based approach that dodges prescriptive standards would be preferable, either as a permanent solution or as a temporary mechanism pending more precise guidance. Overly rigorous regulatory measures can obstruct development, especially in an environment where many enterprises rely on flexible compliance schemes if the process of monitoring FinTech activities is not already refined. It is also important to ensure a level playing field, as the institutional framework often preferentially treats larger institutions at the expense of smaller startups and emerging platforms. Any ex-ante preferences granted to specific organisations must be underpinned by explicit rules, rather than discretionary intervention, to prevent unfair competitive advantages and the confusion that can retard growth. Prior to opting for any exceptional treatment, the authorities should verify the soundness of the proposed concept and confirm that the desired objectives of the policy framework could not be achieved via less intrusive solutions. Such exemptions, when extended, should be qualified by transparent and unequivocal criteria in order to reduce arbitrariness and the risk of sending errant signals of undeserved unfairness.

7.2. Encouraging Public-Private Partnerships

In addition to creating a regulatory environment conducive to development and competition, the formation of public-private partnerships is necessary to realize the potential of financial technology (FinTech) platforms in emerging markets. Public-private partnerships can effectively implement complementary policies that leverage the financial sector's potential to contribute to broad-based sustainable development that is, economic growth combined with increased equality and opportunities for both men and women. An example often mentioned is the Brazilian Development Bank, which was established in the 1950s with the explicit objective of promoting the expansion of small and medium-sized enterprises (SMEs). It financed more than 85 percent of the total amount of credits allocated to SMEs across all economic sectors and all regions. More generally, many countries have development banks that were created exactly to fill the void left by banks that may require a minimum number of guarantees, which SMEs by definition cannot provide (Ivashchenko et al., 2018).

CONCLUSION

The role of FinTech in promoting financial inclusion has gained increasing attention in academia and policymaking circles (Rahman Aleemi et al., 2023). Evidence from emerging markets suggests that FinTech has improved the availability and accessibility of financial services for SMEs, reduced the cost of accessing capital, and fostered financial literacy among businesses. Policymakers should reconsider financial market regulations to create a supportive and competitive environment for sustainable FinTech development, encourage Public-Private Partnerships to consolidate the FinTech ecosystem and promote the widespread adoption of key technological applications, and focus on financial and digital literacy strategies to ready potential FinTech clients to benefit from the dissemination of these services. Financial inclusion (FI) is a key instrument to achieve inclusive growth, mobilising resources, promoting savings and developing payment mechanisms that strengthen financial institutions. A well-inclusive financial sector is essential for achieving many SDGs, such as ending poverty, hunger and reducing inequalities. In addition, G20 nations also recognise FI as a vital pillar of global development. Several factors affect the level of FI. Amongst infrastructure indicators, cell-phone subscriptions promote FI, while rural-urban divides and higher shares of unemployment may affect it adversely. Regarding market structure and competition within banking, countries with broader banking regulations and increased competition coverage tend to have deeper financial penetration. The broader transformation of the financial system over the last three decades was mostly digital. The emergence of Fintech companies (firms applying technological advancement to provide financial services) drastically transformed the banking industry. In the face of competition, banks had to develop digital infrastructure to incorporate Fintech firms in their decision-making process. In addition, a reduction in the overall number of banks pressured the departments to invest more and collaborate more with Fintech.

Financial services are principal enablers of human development and access to finance constitutes one of the most significant impediments to the growth of small and medium-sized enterprises (SMEs) in developing countries. Commercial banks are still the dominant providers of finance for over 90% of formal SMEs, but several large surveys reveal, as well as anecdotal evidence, that banks tend to focus lending on large enterprises and tend to be very reluctant to lend to SMEs. SMEs contribute significantly to the developing countries' economies, but they still remain hugely with untapped potential due to challenges such as the lack of finance. Commercial banks in Zimbabwe have a critical role in the development of SMEs which can be the foundation for economic recovery. Once SMEs formalise they benefit the economy through tax revenues, access to formal credit, increasing investment and profitable lending for commercial banks. For a successful SME sector therefore, the active contribution of SMEs, the financial sector, the central bank and the government is essential, integrating the several interventions such as training, capacity development and sector specific tax benefits. The various initiatives thereby help SMEs to innovate, operate efficiently, reduce the risk of fraud and default and enable banks to lend profitably to SMEs (Sachikonye & Sibanda, 2016). Adopting risk-based KYC and broadening the

range of data available in the credit reference bureau will help reduce this burden for women-owned SMEs. Societal and cultural norms across SSA pose significant barriers for women, especially around household decision-making and economic activities. Addressing these barriers requires engagement from governments, financial institutions, and donors through gender sensitisation training and community-level interventions. Private institutions often misjudge business opportunities in low income markets due to a lack of information, which can be addressed by investing in knowledge products that provide informational data sets. Despite high levels of global funding, many fintech companies struggle to raise capital to grow their balance sheets, so support at the hands of donor organisations via grants, impact investing and guarantees is crucial. Fintech innovation has positively impacted women's financial inclusion by improving household decision-making, financial resilience and income-generating activities, but solutions remain gender blind and are not tailored to women's specific context in SSA.

REFERENCES

Isabel Pavón Cuéllar, L. (2018). International financial inclusion: some multidimensional determinants.
Mhlanga, D. (2024). The role of big data in financial technology toward financial inclusion.
Woldie, A., Mwangaza Laurence, B., & Thomas, B. (2018). Challenges of finance accessibility by SMEs in the democratic republic of Congo: is gender a constraint?.
Rahman Aleemi, A., Javaid, F., & Sajid Hafeez, S. (2023). Finclusion: The nexus of Fintech and financial inclusion against banks' market power
Leong, K. & Sung, A. (2018). FinTech (Financial Technology): What is it and how to use technologies to create business value in FinTech way?.
Mncedi Mahlati, M. (2018). A service delivery framework to unlock the revenue potential of the SME segments of commercercial banks in South Africa.
Elasrag, H. (2016). Islamic finance for SMES.
Yeboah, J., Kwadwo Asirifi, E., & Delali Adigbo, E. (2014). The Role of Banks in Financing Small and Medium Scale Enterprises in Ghana- A Case Study of Universal Banks in Sekondi-Takoradi.
Ivashchenko, A., Britchenko, I., Dyba, M., Polishchuk, Y., Sybirianska, Y., & Vasylyshen, Y. (2018). Fintech platforms in SME's financing: EU experience and ways of their application in Ukraine.
Jun, W. & Qian Ran, X. (2024). Dynamics in digital finance and its impact on SME financing.
Didenko, A. (2018). Regulating FinTech: Lessons from Africa.
Boshkov, T. (2019). Virtual Banking and Financial Inclusion.
Barberis & Ross P. Buckley Douglas W. Arner, J. (2017). FinTech, RegTech, and the Reconceptualization of Financial Regulation.
Sachikonye, S. & Sibanda, M. (2016). An Assessment of SMEs' Financing by Commercial Banks.
Lyons, Angela and Grable, John and Zeng, Ting, Impacts of Financial Literacy on Loan Demand of Financially Excluded Households in China.